Farhan Amin

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Education

Rochester Institute of Technology - Rochester, NY B.S. in Artificial Intelligence, Expected May 2030 (First-year student)

Projects

Music Trainer: Singing & Ear-Training Web App - Personal Project | crazycatz67.github.io/music-trainer

A free, private browser app that teaches singing by drawing your live pitch on a planetarium sky. Built because paid singing apps cost $60–120 a year and hide the raw pitch line.

- Built real-time pitch tracking with the Web Audio API (McLeod pitch method), with the browser's voice-damaging audio filters switched off and checked, and nine practice exercises fitted to each singer's measured range.
- Ran AI models in the browser (HTDemucs source separation, SwiftF0 pitch, ONNX Runtime Web) so any song a user drops in becomes a sing-along, with no audio ever uploaded.
- Rendered an interactive sky of 8,920 real stars and 88 constellations with Three.js, kept under a measured flash limit and under 0.5 ms per frame.
- Led a team of AI coding agents (research, design, build, test, record-keeping) behind a one-command test gate: 859 unit tests and 39 headless-browser test labs with fake microphones; ~35,000 lines of app code.

ASL Fingerspelling Recognizer & Learning Tool - Personal Project | github.com/Crazycatz67/asl-recognizer

A browser web app that reads ASL fingerspelling (A-Z) in real time from a webcam. Inspired by a moment I couldn't communicate with a deaf man in line at a campus event.

- Built a labeled dataset of 10,000+ hand-sign images from Kaggle and other open sources.
- Built the recognition pipeline with MediaPipe for hand tracking and a custom classifier, running fully in the browser with no server.
- Wrote extra logic for the hardest letters: the easily confused ones (M/N, D/O/C) and the moving letters J and Z that need motion tracking.
- Added practice and challenge modes plus a Spell mode that voices signed letters into words. Taught myself the whole stack using Claude Code.

Interactive 3D Hologram Workshop - Personal Project | github.com/Crazycatz67/hologram-workshop

A browser demo that renders a real object, scanned with LiDAR, as a 3D hologram you control with hand gestures. Inspired by the interfaces in Iron Man.

- Scanned a physical object with LiDAR and rendered it in the browser with Three.js.
- Added webcam hand-gesture control to rotate, scale, and measure the object live.

Skills

Languages: JavaScript, Python, HTML/CSS, C++, C#

Tools & Libraries: Web Audio API, Three.js, MediaPipe, ONNX Runtime Web, Node.js test runner, headless Chrome testing, Git/GitHub Pages, Claude Code, 3D printing and modeling

Approach: AI-assisted development: directing a team of AI agents with specs, code review and automated test gates. Learned to code and use AI tools from the ground up by designing and building real projects.

Experience

Independent AI-Assisted Development - Self-Directed

Taught myself software development and AI tools with no formal instruction. Designed, built, tested and published three working web apps (ASL recognizer, 3D hologram workshop, music trainer) in my first two months of college, learning each technology as I went.

3D Printing & AI Workflow Internship - Remote

Remote internship working with 3D printing and learning to use AI tools efficiently.

Harkness Independent Study - High School (Grades 11–12)

Built websites and games in HTML, JavaScript, Python, C++ and C# through a self-directed Harkness program.